The SD-Nonfriendly Converse for Arbitrary c_0-Sums

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Status and source question

This packet gives a likely valid partial answer to the question following Proposition 4.3(c) of Bilik–Kadets–Shvidkoy–Sirotkin–Werner [1]. For a compact Hausdorff space K, a Banach space E, and an arbitrary range space W, that proposition says

$$\big[\text{every } T \in \mathrm{SD}(C(K,E),W) \text{ is } C\text{-narrow}\big] \quad \Longrightarrow \quad \mathrm{SD}(E)=\{0\}.$$

The authors ask whether the converse holds. Their May 2025 monograph [2] repeats the result as Proposition 8.3.4 and still lists the converse as Question 8.2.

We do not settle the unrestricted converse. We prove it for every Banach space assembled as an arbitrary c_0-sum of separable USD-nonfriendly spaces. The resulting space need not be separable. The proof is an M-summand permanence argument which extends the isolated $E = c_0$ observation in the source.

Notation and source facts

Write $\mathsf{P}_K(E)$ for the following property:

> For every Banach space W, every strong Daugavet operator $T : C(K,E) \to W$ is C-narrow.

Write $\mathsf{P}(E)$ if $\mathsf{P}_K(E)$ holds for every compact Hausdorff K.

We use four results proved in [1] (also reproduced in Chapter 8 of [2]).

1. If $X = X_1 \oplus_\infty X_2$ and $T \in \mathrm{SD}(X,W)$, then each restriction $T|_{X_j}$ is a strong Daugavet operator.
2. If $T : C(K,E) \to W$, then T is C-narrow if and only if, for every $x \in E$,
$$T_x : C(K) \to W, \qquad T_x g = T(g \otimes x),$$
is C-narrow.
3. Finite sums of scalar C-narrow operators on $C(K)$ are C-narrow.
4. Every separable USD-nonfriendly space has $\mathsf{P}(E)$, and every USD-nonfriendly space is SD-nonfriendly.

Recall that $E_0 \subset E$ is an M-summand when $E = E_0 \oplus_\infty F$ for some closed $F \subset E$.

Two elementary closure lemmas

Lemma 1 (Norm closure)**.** *For fixed K and W, the scalar C-narrow operators $C(K) \to W$ form a norm-closed class.*

Proof. Let $R_n \to R$ in operator norm, with every R_n C-narrow. Use the local unit-bump characterization from [1]. Given a nonempty open $U \subset K$ and $\varepsilon > 0$, choose n with $\|R - R_n\| < \varepsilon/2$. There is a nonnegative $g \in C(K)$ with $\|g\| = 1$, $\operatorname{supp} g \subset U$, and $\|R_n g\| < \varepsilon/2$. Then $\|Rg\| < \varepsilon$. Thus R is C-narrow. □

Lemma 2 (Finite ℓ_∞-stability)**.** *If $\mathsf{P}_K(E_j)$ holds for $j = 1, \dots, m$, then it holds for $E = E_1 \oplus_\infty \cdots \oplus_\infty E_m$.*

Proof. Let $T : C(K, E) \to W$ be strong Daugavet. Since

$$C(K, E) = C(K, E_1) \oplus_\infty \cdots \oplus_\infty C(K, E_m),$$

the restriction $T_j = T|_{C(K,E_j)}$ is strong Daugavet, hence C-narrow by $\mathsf{P}_K(E_j)$.

For $x = (x_1, \dots, x_m) \in E$,

$$T_x g = \sum_{j=1}^{m} T_j(g \otimes x_j).$$

Each summand is a scalar C-narrow operator, and their finite sum is C-narrow. Therefore every T_x is C-narrow, so the scalar restriction criterion gives that T is C-narrow. □

Dense M-summand permanence

Theorem 1 (Dense M-summand principle)**.** *Let E have a family $(E_\alpha)_{\alpha \in A}$ of M-summands whose union is norm dense in E. If every E_α has $\mathsf{P}_K(E_\alpha)$, then E has $\mathsf{P}_K(E)$.*

Proof. Let $T : C(K, E) \to W$ be strong Daugavet. If $E = E_\alpha \oplus_\infty F_\alpha$, then pointwise decomposition gives

$$C(K, E) = C(K, E_\alpha) \oplus_\infty C(K, F_\alpha).$$

Hence $T|_{C(K,E_\alpha)}$ is strong Daugavet and, by hypothesis, C-narrow. Consequently T_x is scalar C-narrow for every $x \in \bigcup_\alpha E_\alpha$.

For all $x, y \in E$,

$$\|T_x - T_y\| \le \|T\| \, \|x - y\|.$$

The union of the E_α's is dense, so Lemma 1 implies that T_x is C-narrow for every $x \in E$. The scalar restriction criterion now shows that T is C-narrow. □

There is a parallel observation for SD-nonfriendliness.

Lemma 3 (Dense SD-nonfriendly M-summands)**.** *Under the hypotheses of Theorem 1, if every E_α is SD-nonfriendly, then E is SD-nonfriendly.*

Proof. If $S : E \to Y$ is strong Daugavet, then every restriction $S|_{E_\alpha}$ is strong Daugavet and hence zero. Density of the union and continuity of S give $S = 0$. □

The c_0-sum theorem

Theorem 2 (Arbitrary c_0-sums)**.** *Let $(E_\gamma)_{\gamma\in\Gamma}$ be any family of Banach spaces and put*

$$E = \Big(\bigoplus_{\gamma\in\Gamma} E_\gamma\Big)_{c_0}.$$

Then:

1. *if every E_γ has $\mathsf{P}_K(E_\gamma)$, then E has $\mathsf{P}_K(E)$;*
2. *if every E_γ is SD-nonfriendly, then E is SD-nonfriendly.*

The index set Γ is unrestricted.

Proof. For a finite $F \subset \Gamma$, let

$$E_F = \bigoplus_{\gamma\in F}^{\ell_\infty} E_\gamma.$$

It is an M-summand of E, and the union of all E_F is the finitely supported subspace, which is norm dense in the c_0-sum. Lemma 2 gives $\mathsf{P}_K(E_F)$, so Theorem 1 proves (1). For (2), repeated restriction to the coordinate M-summands shows that every finite E_F is SD-nonfriendly; Lemma 3 then applies. $\square$

Corollary 1 (A broad positive class for the source question)**.** *If every E_γ is separable and USD-nonfriendly, then*

$$E = \Big(\bigoplus_{\gamma\in\Gamma} E_\gamma\Big)_{c_0}$$

is SD-nonfriendly and has $\mathsf{P}(E)$. In particular, this holds when every E_γ is finite-dimensional.

Proof. Each component is SD-nonfriendly and has $\mathsf{P}(E_\gamma)$ by the source results. Apply Theorem 2 for every compact K. $\square$

Thus Proposition 4.3(c) is an equivalence throughout this class, including nonseparable examples obtained from uncountable Γ.

Why the full converse is not claimed

The proof uses a feature absent from a general SD-nonfriendly space: every vector is norm approximable inside M-summands on which a strong Daugavet operator restricts as a strong Daugavet operator. A general separable subspace does not have this restriction property.

Two natural full-solution routes stop at the same point. First, quotienting the scalar maps $T_x : g \mapsto T(g \otimes x)$ by the space of C-narrow operators would settle the question if the induced operator on E were strong Daugavet. A strong Daugavet witness only controls Tf, however, and does not control $T(gf)$ for scalar cutoffs g. Second, localization and ultrapower arguments need exactly the same cutoff stability to turn vector-valued witnesses into approximately strong Daugavet maps on E. No such implication follows from the definitions.

The packet therefore promotes Corollary 1 as a partial result and leaves the unrestricted Question 8.2 open.

Novelty and review notes

The cheap run indexes were searched for the exact arXiv id and for the terms SD-nonfriendly, USD-nonfriendly, C-narrow, strong Daugavet, M-summand, direct sum, and c_0-sum. A bounded primary-source web search found [1] and the 2025 monograph [2], but no later answer or explicit version of Theorems 1–2.

Human review should focus on the use of the strong-Daugavet restriction theorem for ℓ_∞-summands and on the scalar-restriction characterization of C-narrowness. Once those source results are granted, the remaining argument is elementary.

References

[1] D. Bilik, V. Kadets, R. Shvidkoy, G. Sirotkin, and D. Werner, *Narrow operators on vector-valued sup-normed spaces*, arXiv:math/0106227, 2001.

[2] V. Kadets, M. Martín, A. Rueda Zoca, and D. Werner, *Banach spaces with the Daugavet property*, preliminary monograph, May 2025, especially Proposition 8.3.4 and Question 8.2.